FlareVM - Arsenal of Tools

FlareVM, or "**Forensics, Logic Analysis, and Reverse Engineering**," stands out as a comprehensive and carefully curated collection of specialised tools uniquely designed to meet the specific needs of reverse engineers, malware analysts, incident responders, forensic investigators, and penetration testers

Arsenal of Tools

Reverse Engineering & Debugging

Reverse engineering is like solving a puzzle backward: you take a finished product apart to understand how it works. Debugging is identifying errors, understanding why they happen, and correcting the code to prevent them.

- **Ghidra** - NSA-developed open-source reverse engineering suite.
- **x64dbg** - Open-source debugger for binaries in x64 and x32 formats.
- **OllyDbg** - Debugger for reverse engineering at the assembly level.
- **Radare2** - A sophisticated open-source platform for reverse engineering.
- **Binary Ninja** - A tool for disassembling and decompiling binaries.
- **PEiD** - Packer, cryptor, and compiler detection tool.

Disassemblers & Decompilers

Disassemblers and Decompilers are crucial tools in malware analysis. They help analysts understand malicious software's behaviour, logic, and control flow by breaking it into a more understandable format. The tools mentioned below are commonly used in this category.

- **CFF Explorer** - A PE editor designed to analyze and edit Portable Executable (PE) files.
- **Hopper Disassembler** - A Debugger, disassembler, and decompiler.
- **RetDec** - Open-source decompiler for machine code.

Static & Dynamic Analysis

Static and dynamic analysis are two crucial methods in cyber security for examining malware. Static analysis involves inspecting the code without executing it, while dynamic analysis involves observing its behaviour as it runs. The tools mentioned below are commonly used in this category.

- **Process Hacker** - Sophisticated memory editor and process watcher.
- **PEview** - A portable executable (PE) file viewer for analysis.

- **Dependency Walker** - A tool for displaying an executable's DLL dependencies.
- **DIE (Detect It Easy)** - A packer, compiler, and cryptor detection tool.

Forensics & Incident Response

Digital Forensics involves the collection, analysis, and preservation of digital evidence from various sources like computers, networks, and storage devices. At the same time, Incident Response focuses on the detection, containment, eradication, and recovery from cyberattacks. The tools mentioned below are commonly used in this category.

- **Volatility** - RAM dump analysis framework for memory forensics.
- **Rekall** - Framework for memory forensics in incident response.
- **FTK Imager** - Disc image acquisition and analysis tools for forensic use.

File Analysis

File Analysis is a technique used to examine files for potential security threats and ensure proper file permissions.

- **FileInsight** - A program for looking through and editing binary files.
- **Hex Fiend** - Hex editor that is light and quick.
- **HxD** - Binary file viewing and editing with a hex editor.

Scripting & Automation

Scripting and Automation involve using scripts such as PowerShell and Python to automate repetitive tasks and processes, making them more efficient and less prone to human error.

- **Python** - Mainly automation-focused on Python modules and tools.
- **PowerShell Empire** - Framework for PowerShell post-exploitation.

Sysinternals Suite

The Sysinternals Suite is a collection of advanced system utilities designed to help IT professionals and developers manage, troubleshoot, and diagnose Windows systems.

- **Autoruns** - Shows what executables are configured to run during system boot-up.
- **Process Explorer** - Provides information about running processes.
- **Process Monitor** -Monitors and logs real-time process/thread activity.

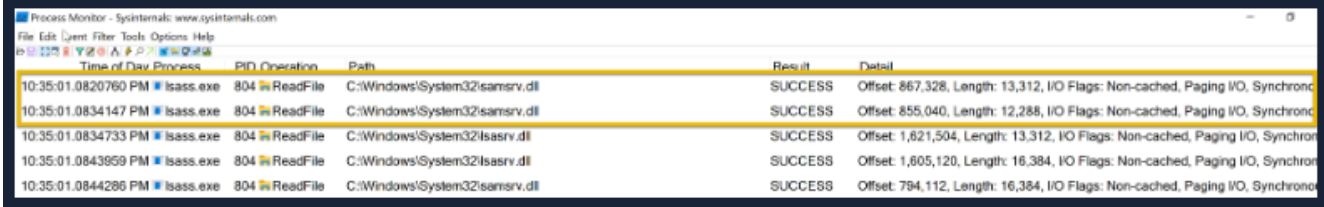
Commonly Used Tools for Investigation: Overview

Tool	Investigative Value
Procmon	A helpful tool for tracking system activity, especially regarding malware research, troubleshooting, and forensic investigations.
Process Explorer	Allows you to see the Process of the Parent-child relationship, DLLs loaded, and its path.
HxD	Malicious files can be examined or altered via hex editing.
Wireshark	Observing and investigating network traffic to look for unusual activity.
CFF Explorer	Can generate file hashes for integrity verification, authenticate the source of system files, and validate their validity.
PEStudio	Static analysis or studying executable file properties without running the files.
FLOSS	Extracts and de-obfuscates all strings from malware programs using advanced static analysis techniques.

Process Monitor (Procmon)

It lets you **see, record, and keep track of system and Windows file activity in real-time**. Process Monitor is helpful for **tracking system activity, especially regarding malware research, troubleshooting, and forensic investigations**.

Here's how to utilize it effectively for investigation.

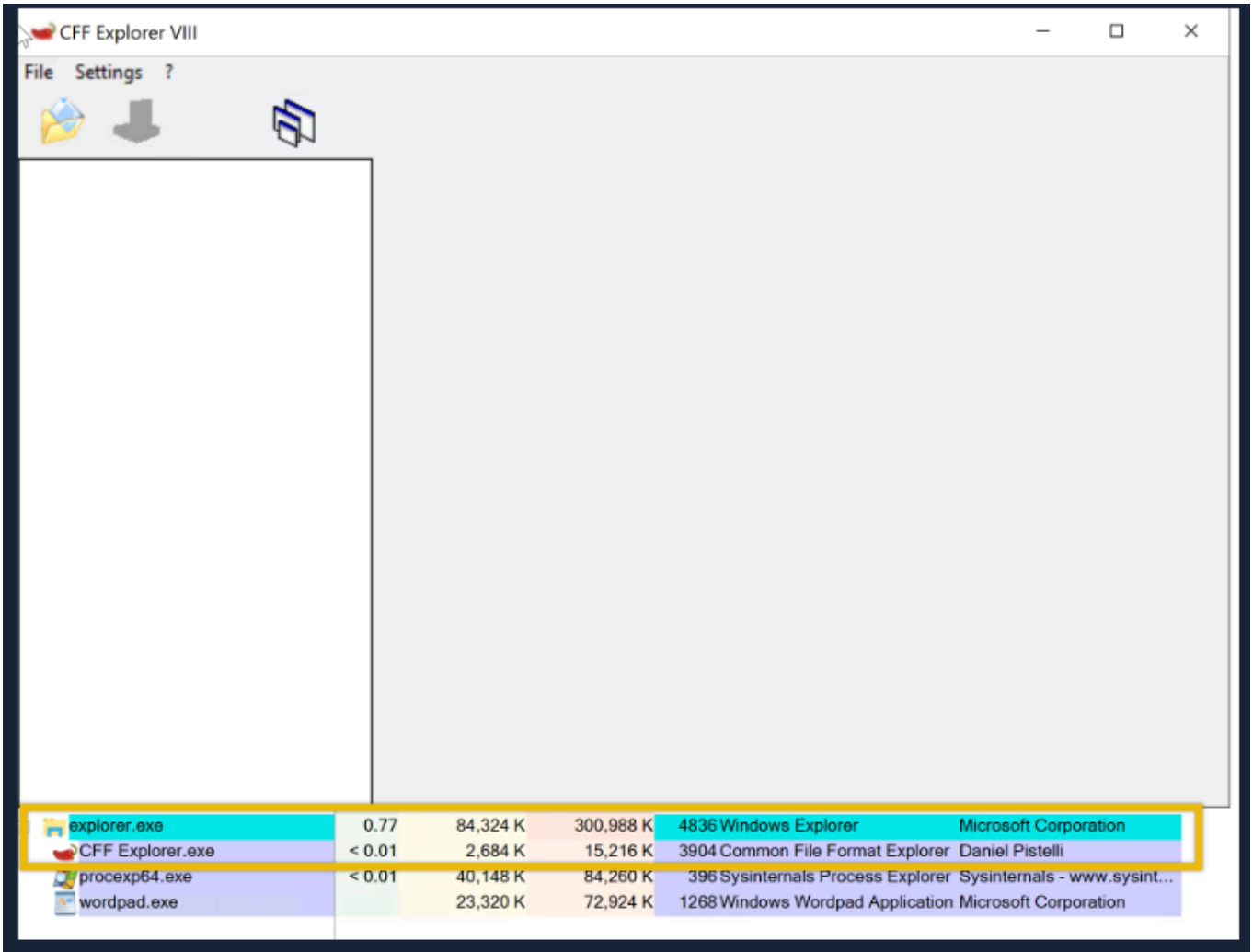


According to this log entry, the Local Security Authority Subsystem Service (LSASS)- related process lsass.exe has successfully read a file. LSASS handles authentication and frequently communicates with crucial system files such as lsasrv.dll (Local Security Authority Server Service).

Although this is a standard system process, LSASS may be the target of credential dumping attacks if you are examining logs for indications of malicious activity. Mimikatz and other tools frequently try to access LSASS memory. In these situations, you should watch for any additional suspicious activity related to LSASS, such as odd access patterns or processes reading or writing to lsass.exe.

Process Explorer (Procexp)

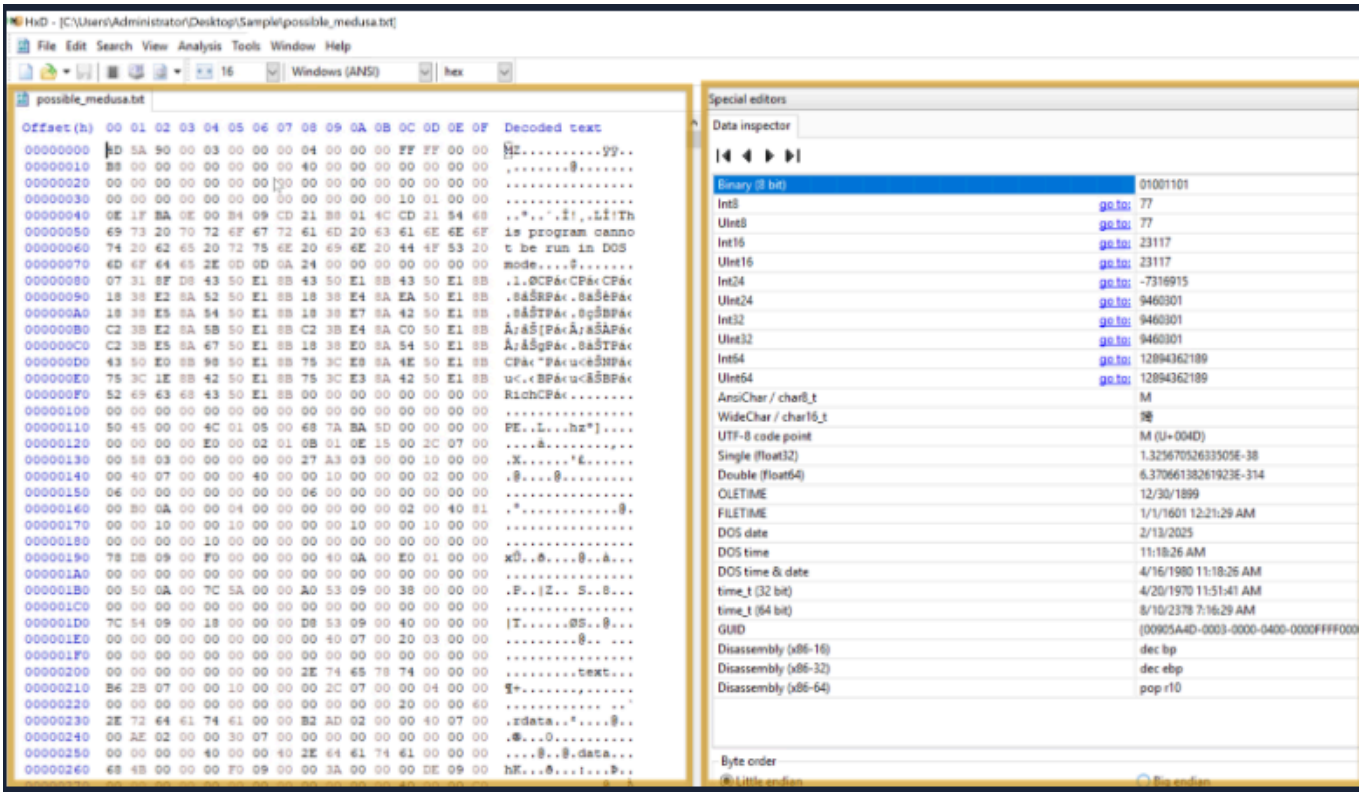
in-depth insights into the active processes running on your computer. It allows you to delve into the inner workings of your system, providing a comprehensive list of currently running processes and their linked user accounts.



As you can see from the image above, the CFF Explorer app is open. Using **Process Explorer(procexp),** located on the desktop, we identified the process and its parent process. This is usually pretty useful when we want to monitor what other processes are being spawned, such as from a Word document, an LNK file, or even an ISO file, as threat actors typically abuse these.

HxD

hex editor for editing files, memory, and drives of any capacity. It can be applied to forensic investigation, data recovery, debugging, and exact manipulation of binary data



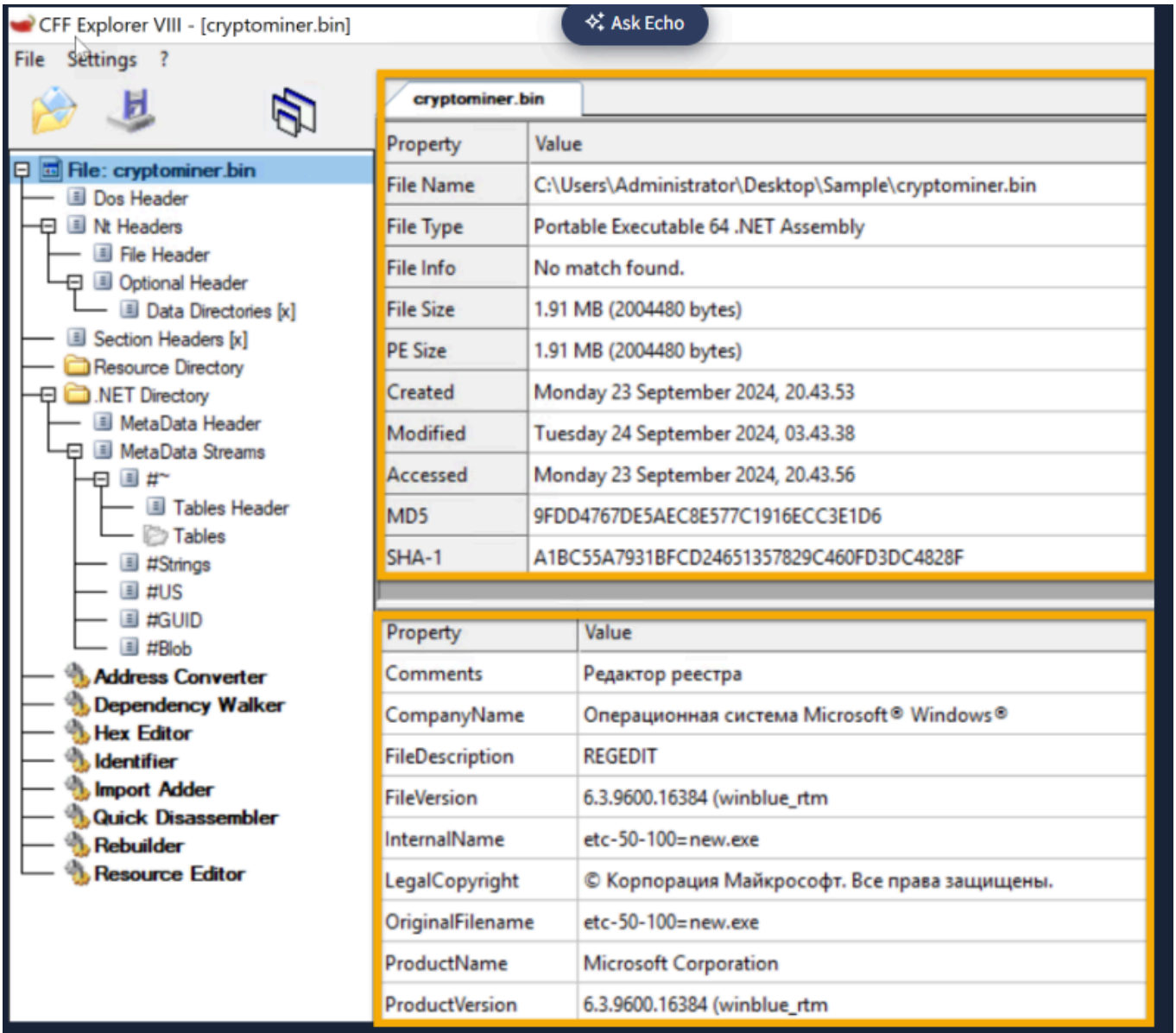
This HxD hex editor snapshot shows the binary file **possible_medusa.txt**. The hex data on the left indicates the file's contents in hexadecimal, and the ASCII interpretation appears on the right. Interestingly, the file starts with **4D 5A (Little Endian)**, indicating it is executable.

The **Data Inspector** on the right allows you to examine individual bytes by displaying their values in many data types (e.g., integer, float), facilitating a more straightforward data evaluation.

By permitting in-depth examination of a file's unprocessed hexadecimal data, HxD facilitates inquiry by identifying file kinds, structures, and possible corruption. Its Data Inspector feature helps by offering insights into particular byte values.

CFF Explorer

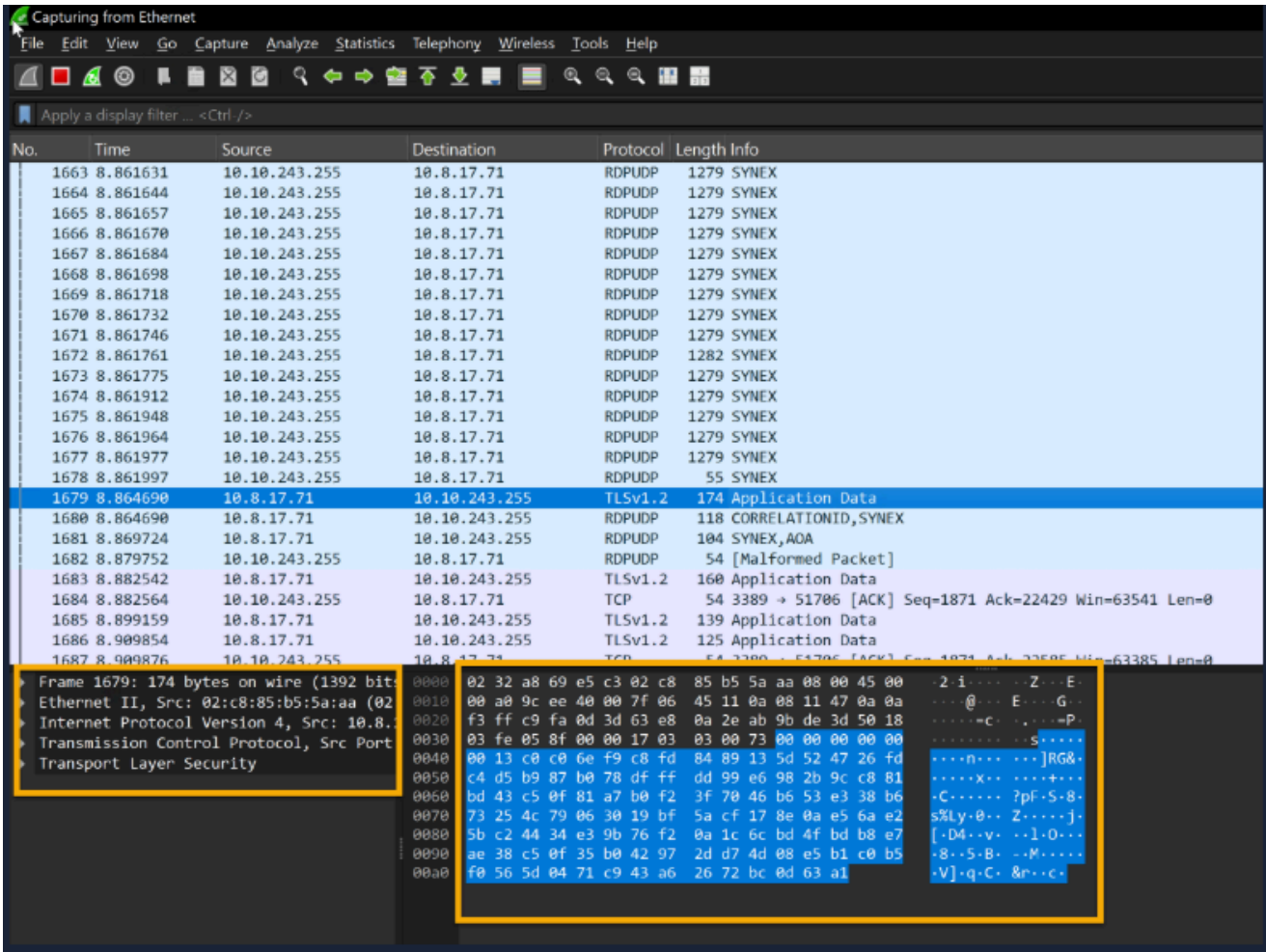
CFF Explorer's comprehensive file information, investigators can generate file hashes for integrity verification, authenticate the source of system files, and validate their validity (e.g., by looking for unusual alterations)



The **cryptominer.bin** file's details are displayed in the sample. On September 23, 2024, a 64-bit Portable Executable file was generated. The file's information can be verified by its hashes (SHA-1 and MD5). During investigations, this tool aids in confirming file information lookup and locating possible problems.

Wireshark

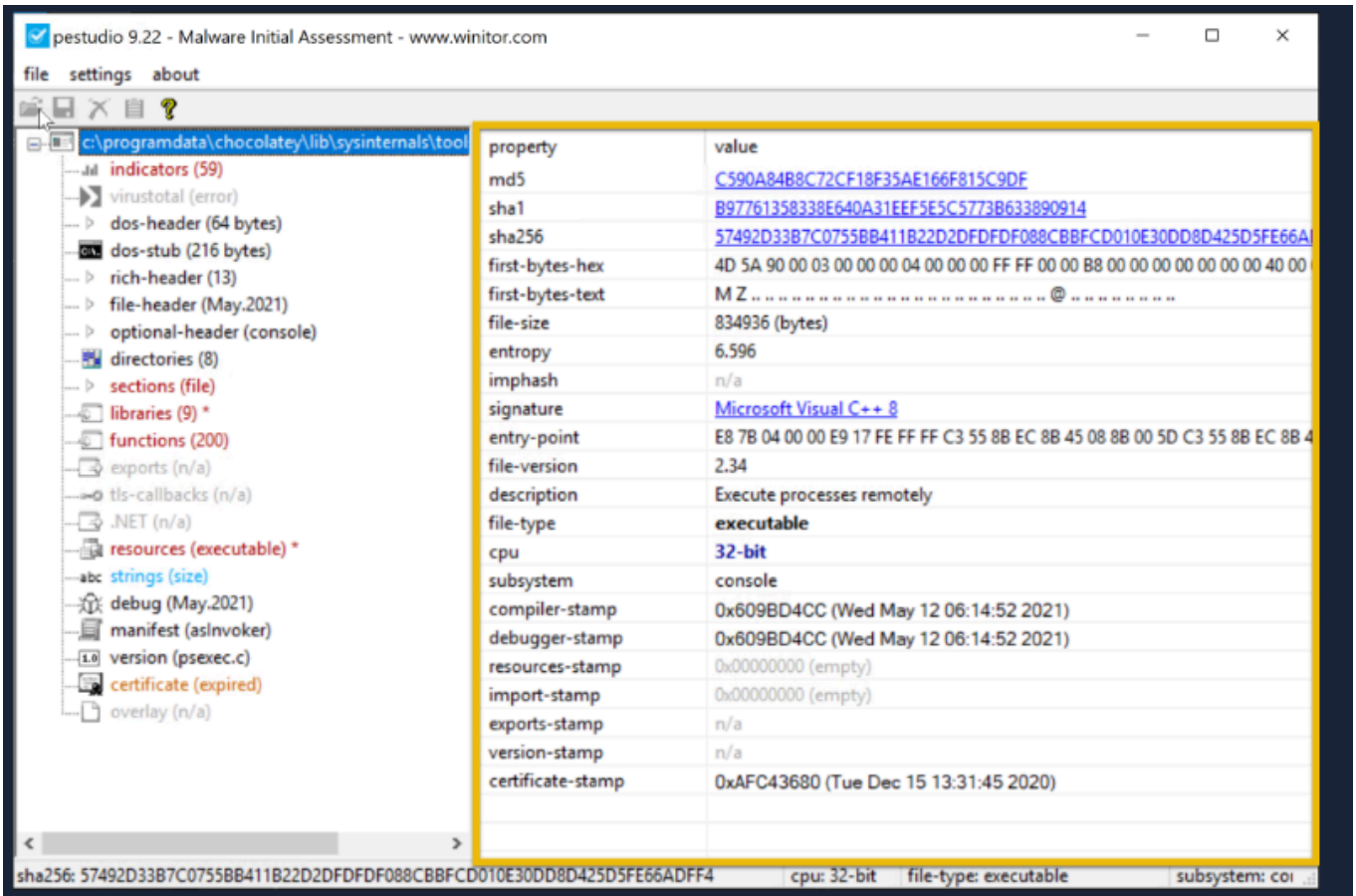
Wireshark is a powerful tool that investigators may use to hunt down dubious connections, examine protocols, and spot possible assaults or data exfiltration



This Shows captured packets with details about the protocol, source, destination, and other information. Most packets show that TLSv1.2 and TCP are being used for encrypted transmission. The raw packet data is shown in ASCII and hexadecimal forms, with a significant chunk encrypted using TLSv1.2.

PEStudio

Static analysis, or studying executable file properties without running the files, is done with PEstudio.



This example shows the examination of an executable file, PSexec.exe (**not in the VM; this is a pure example only**), using PEstudio 9.22, a static malware analysis tool. The file has a dual purpose—legitimate for system administrators but potentially exploited by hackers for remote access.

The file's **entropy value of 6.596** indicates a remote chance of packing or encryption, which is typical of dangerous software. **Version 2.34** of this 32-bit console-based application allows it to run programs remotely, a feature frequently used to migrate laterally during attacks. The file is assembled using **Visual C++ 8**.

The dual-use nature of **PsExec, typically legitimate but suspicious in compromised environments, combined with low to medium indicators and moderately high entropy, makes its presence on a system concerning, especially if remote code execution is not expected. Its use in post-exploitation phases warrants further investigation to determine if it's being misused maliciously.**

FLOSS

Using advanced static analysis techniques, the FLARE Obfuscated String Solver (FLOSS, formerly FireEye Labs Obfuscated String Solver) automatically extracts and de-obfuscates all strings from malware programs.

```
Terminal

PS C:\Users\Administrator\Desktop\Sample > floss .\cobaltstrike.exe
INFO: floss: extracting static strings
finding decoding function features: 100%|
INFO: floss.stackstrings: extracting stackstrings from 50 functions
extracting stackstrings: 100%|
INFO: floss.tightstrings: extracting tightstrings from 4 functions...
extracting tightstrings from function 0x402e80: 100%|
INFO: floss.string_decoder: decoding strings
emulating function 0x402e80 (call 1/1): 100%|
INFO: floss: finished execution after 265.61 seconds
INFO: floss: rendering results
```

▼ View Results

```
Terminal

FLARE FLOSS RESULTS (version v3.1.0-0-gdb9af41)

+------------------------+------------------------------------------------------------
| file path              | cobaltstrike.exe
| identified language    | unknown
| extracted strings      |
|  static strings        | 189 (2050 characters)
|   language strings     |   0 (   0 characters)
|  stack strings         | 0
|  tight strings         | 0
|  decoded strings       | 0
+------------------------+------------------------------------------------------------
```

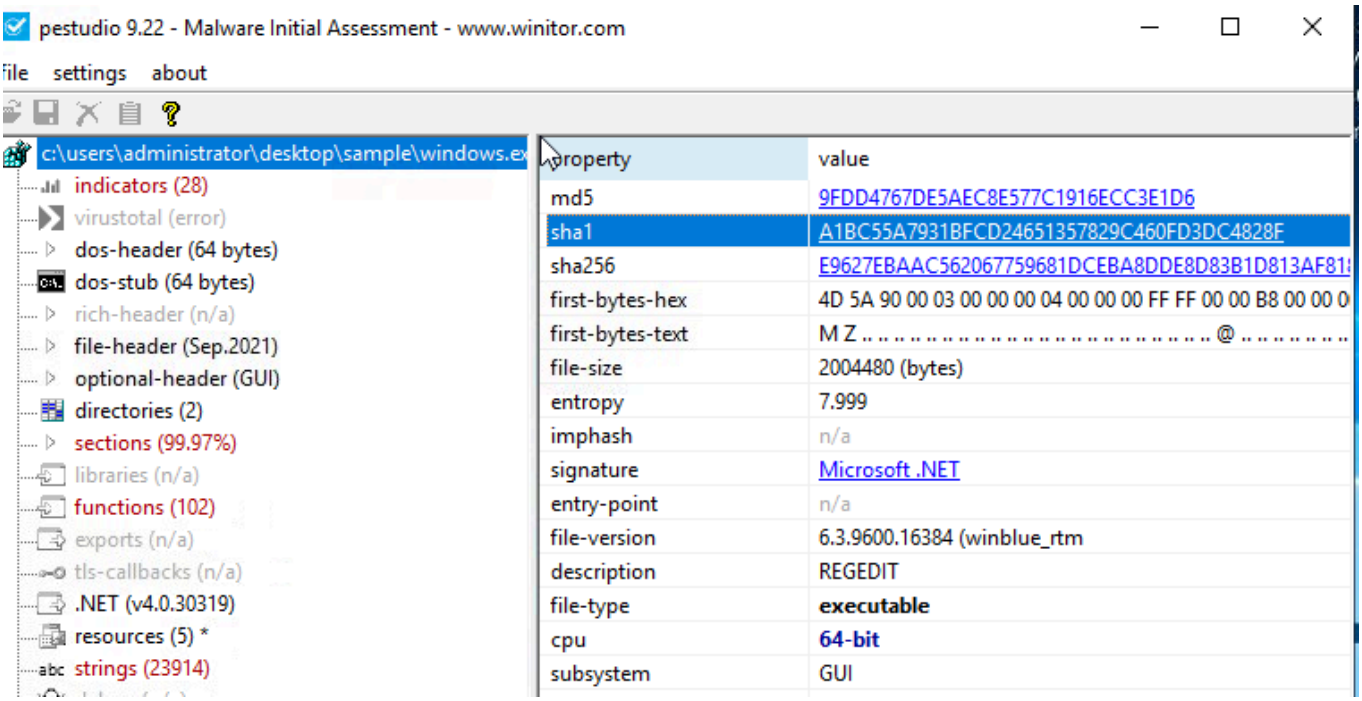
In the example above, FLOSS extracted **189 static strings** from the binary, which may contain hardcoded information such as file **paths**, **URLs** (likely for command-and-control servers), **IP addresses**, **API calls**, **error messages**, **registry**, **encryption keys**, and **configuration data**. However, no decoded strings were identified, suggesting that FLOSS did not detect or decode dynamically generated or obfuscated strings during this analysis. Malware frequently uses obfuscated strings to conceal its malicious behavior.

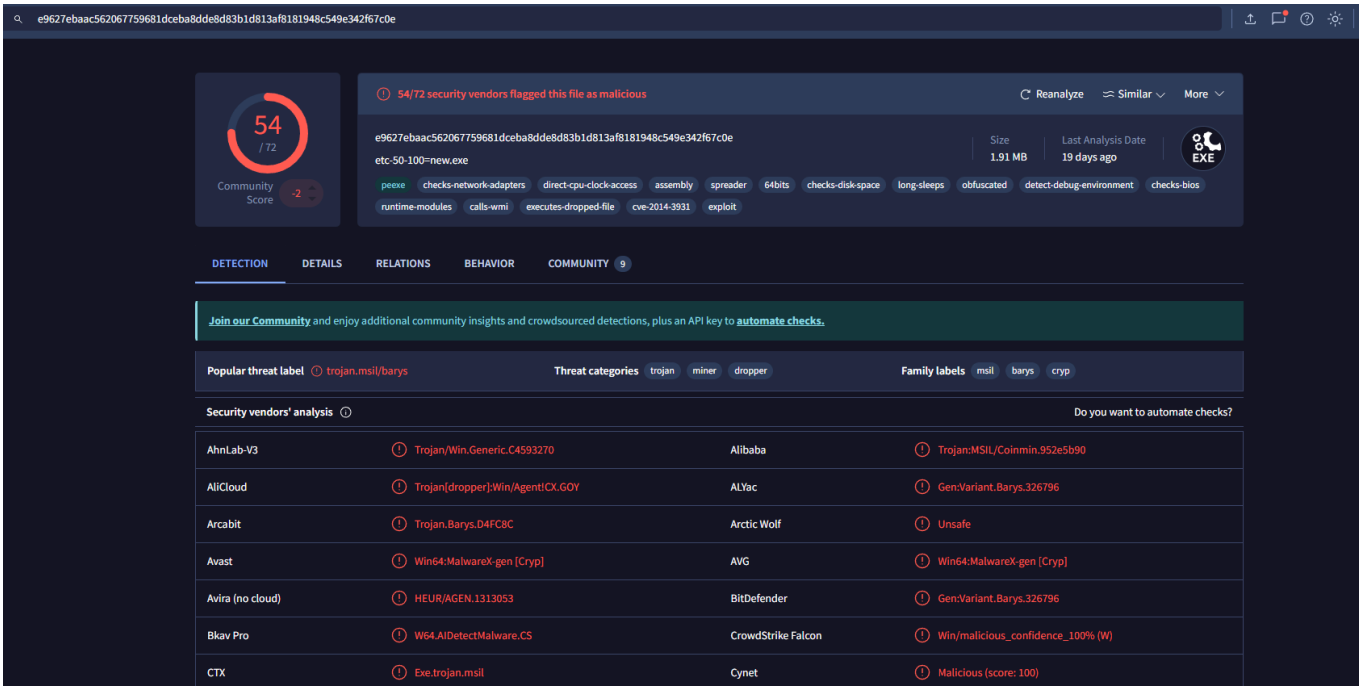
Analysing Malicious Files!

A suspicious **windows.exe** file was downloaded by a user on 09/24/2024 at 3:43 AM. This download was flagged as a potential threat. The monitoring team has sent you an email requesting to perform an analysis of it. They have sent you the file, which is now in the **C:\Users\Administrator\Desktop\Sample** folder.

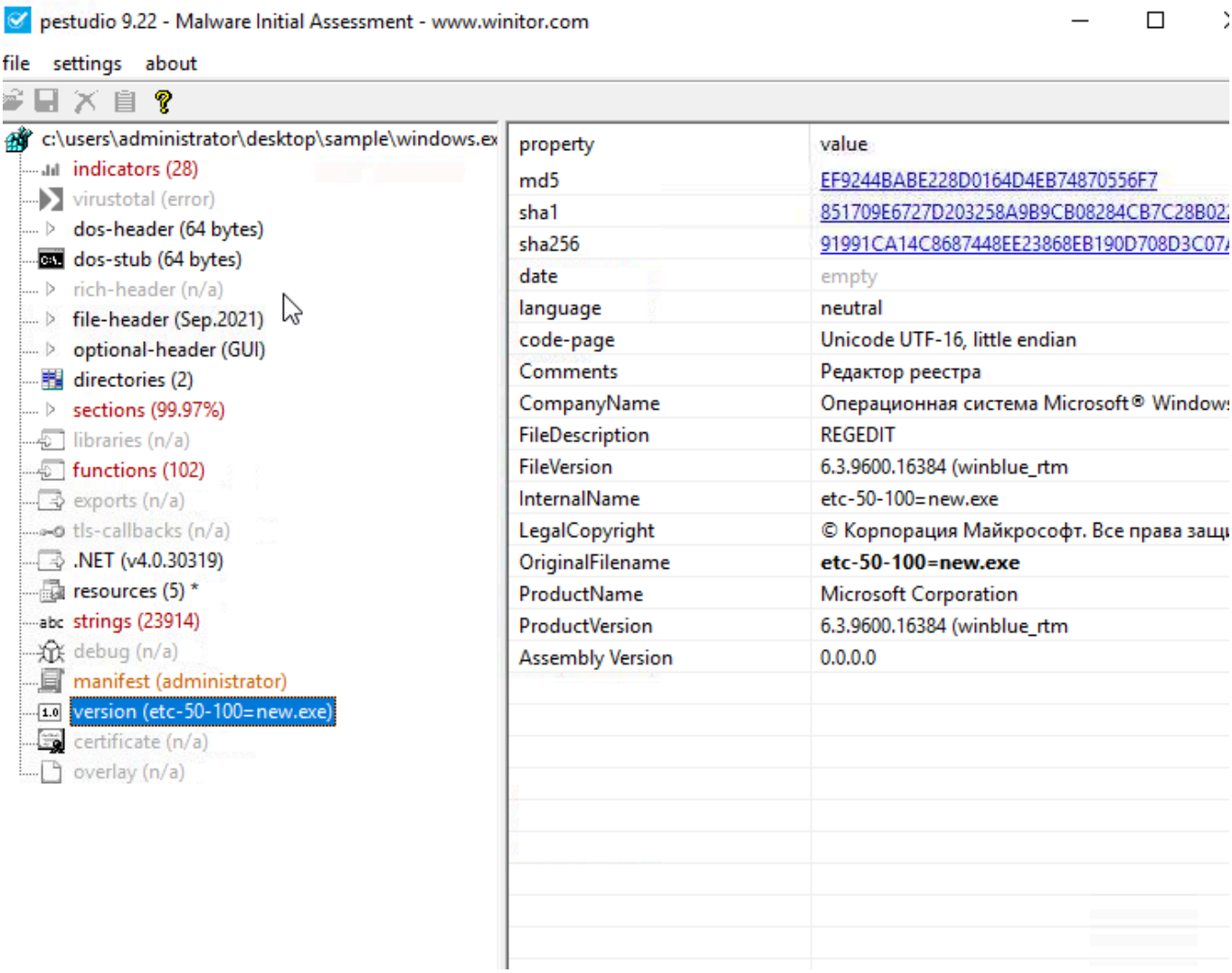
Analysis using PEStudio

i opened the file in PEStudio and took the MD5 and SHA-1 hashes and then put them into compare these hashes on an established database like virus total to look for known malware detections.

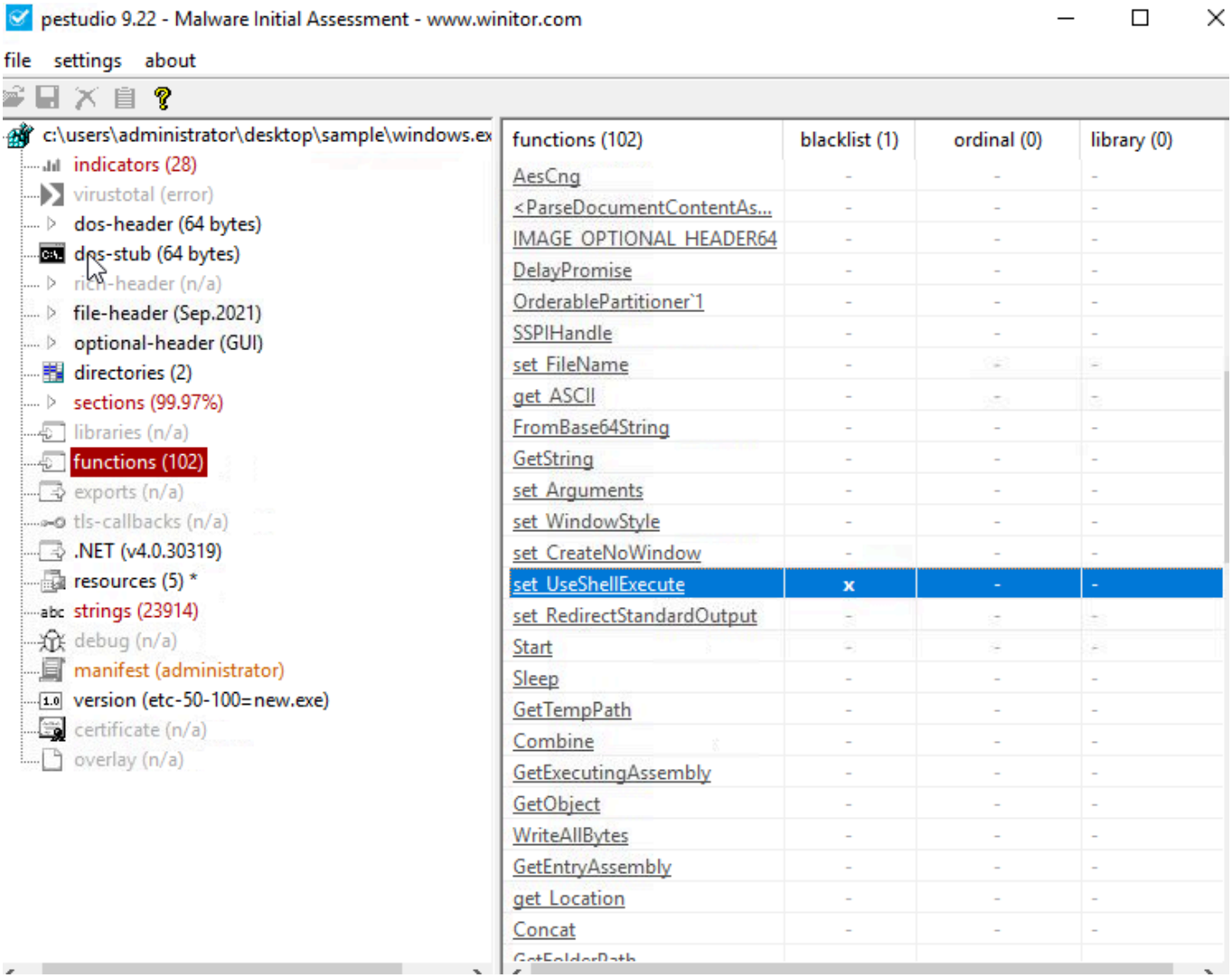




also in the version tab it shows that file isnt exactly what its claiming to be as its comments and company name is in Russian



The absence of a **rich header** indicates that the file is potentially packed or obfuscated to avoid detection by static analysis tools. This is typical behaviour of sophisticated malware that tries to evade detection by altering critical sections of its PE file.



`set_UseShellExecute` : This function allows the process to use the operating system's shell to execute other processes. This is often seen in malware that spawns additional processes to carry out malicious actions.

`CryptoStream, RijndaelManaged, CipherMode, CreateDecryptor` : These APIs indicate that the executable uses cryptographic functions, specifically Rijndael (AES encryption). Malware may use cryptography to encrypt communication and files or even implement ransomware functionality.

Analysis using FLOSS

I used floss on windows to extract contents and put it on a txt file

```
PS C:\Users\Administrator\Desktop\Sample > Floss.exe .\windows.exe > windows.txt
WARNING: floss: .NET language-specific string extraction is not supported yet
WARNING: floss: FLOSS does NOT attempt to deobfuscate any strings from .NET binaries
INFO: floss: disabled string deobfuscation
INFO: floss: extracting static strings
INFO: floss: finished execution after 0.89 seconds
INFO: floss: rendering results
FLARE-VM 02/28/2026 04:23:00
PS C:\Users\Administrator\Desktop\Sample >
```

windows.txt - Notepad

File Edit Format View Help

```
set_WorkingDirectory
Exception
MemoryStream
CryptoStream
System.Security.Cryptography
RijndaelManaged
SymmetricAlgorithm
set_KeySize
set_Mode
CipherMode
GetBytes
Rfc2898DeriveBytes
DeriveBytes
CreateDecryptor
ICryptoTransform
Stream
CryptoStreamMode
Write
Close
IDisposable
Dispose
ToArray
REGEDIT
 Microsoft
 Windows
Microsoft Corporation
```

```
6.3.9600.16384 (winblue_rtm
$131e14cc-f38c-4324-a4b4-f9f47d8f4690
WrapNonExceptionThrows
ffff
`ff`
`fof
ffff
<?xml version="1.0" encoding="utf-8"?>
<assembly manifestVersion="1.0" xmlns="urn
```